

Cognitive Load Estimation from Eye-Tracking Data via Cross-Domain & Test-Time Adaptation

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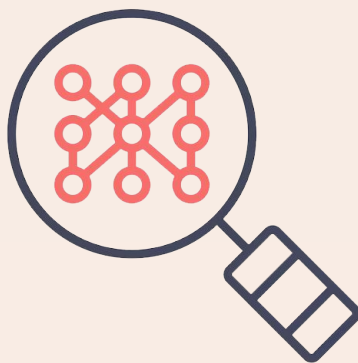
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Demo Notebook URL: [T2430468](https://colab.research.google.com/github/ahmedlamha/CL-ET-CDTA/blob/main/CL-ET-CDTA.ipynb)



Background:



Cognitive Load Theory

- A quantitative measure of **mental effort** and **capacity** (Swaller, 2024)
- Constrained by the **limited working memory**
- Varies across individuals based on prior knowledge & personal cognitive abilities



Eye-tracking for CL Detection

- Proven as a **standalone**, efficient & non-obstructive alternative method to EEG (Aygün et al., 2022).
- EEG data provided no significant performance improvement where models based solely on eye-tracking achieved a **higher accuracy**.
- Pupillometry data provides real-time indicators of **mental strain**

Research Rationale:

Problem Statement

Current eye-tracking CL classifiers face a critical generalization gap, where **inter-subject variance causes drastic accuracy degradation** on new users. As of now, there is a lack of a unified framework that bridges the gap between source & target domains while also handling live fluctuations in estimating CL via eye-tracking data.

The Need to address the Domain Shift

Without adaptation, **performance of static models degrades** when transferred to unseen data due to statistical feature shifts.

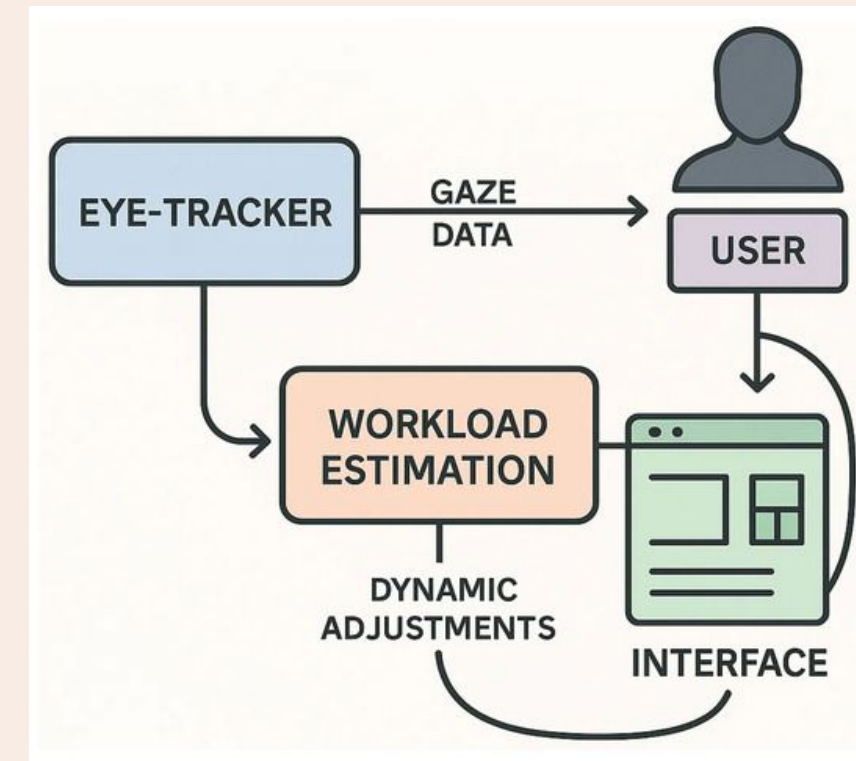
Cross-Domain & Test-Time Adaptation

mathematically align distributions in real-time without human intervention.

Through **Self-Supervision, Pseudo-Labels** can be generated from high-confidence predictions & be used to self-calibrate, removing the need for ground-truth inputs (Xiao & Snoek, 2024).

Implications & Outcomes in HCI:

- Examples of the Applications of CL Estimation: Air Traffic Control (ATC) & Aviation, Driver Distraction Monitoring System, Measuring Learning Effectiveness in Virtual Classrooms, Dynamic UI Adaptation in **SIID**-Aware Assistive Systems.
- **Cross-Domain Adaptation** can address *global* inter-subject **physiological** variabilities.
- **Test-Time Adaptation** can address *local* instance-specific variations such as lighting changes, sensor dropouts & changes in hardware setups.



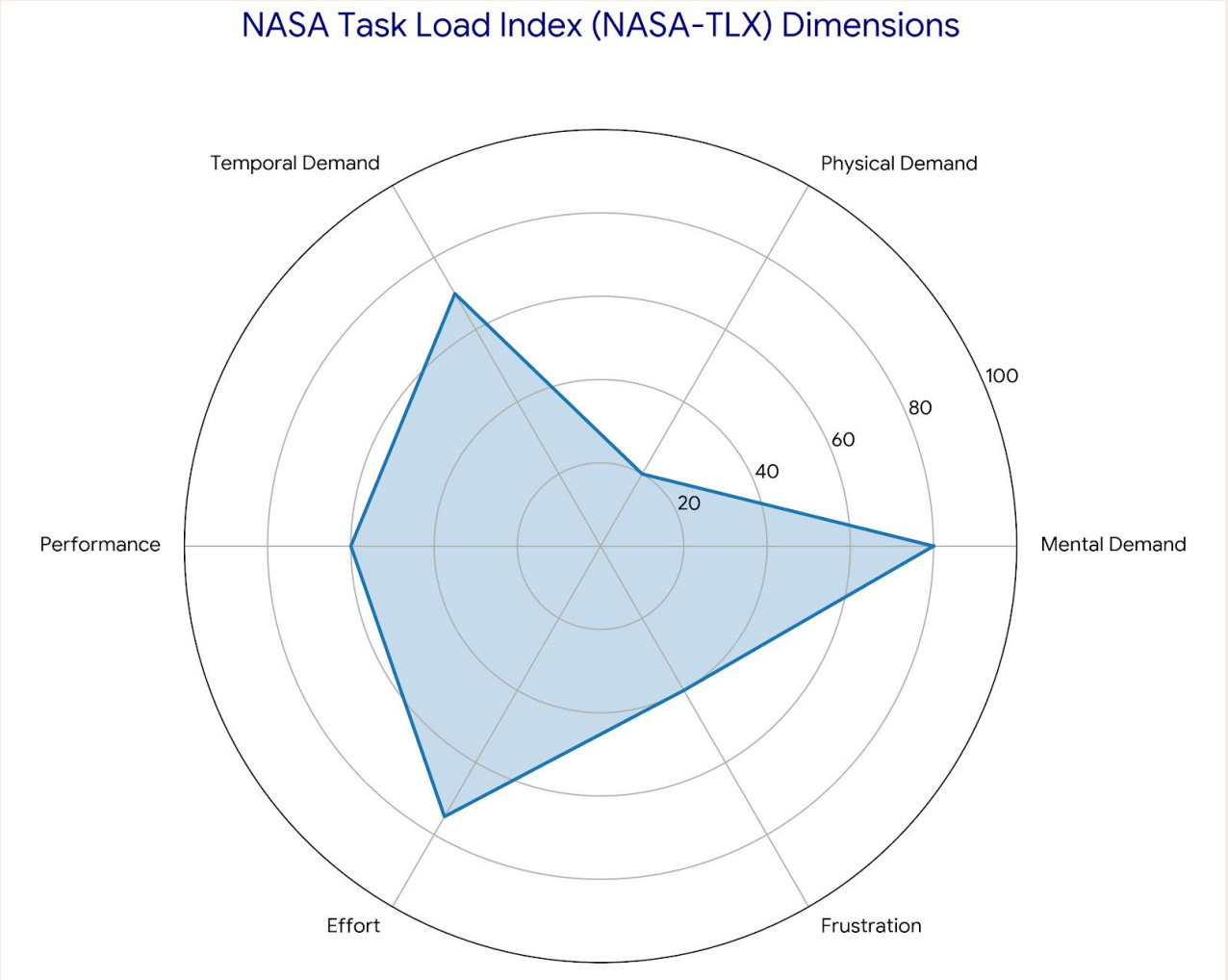
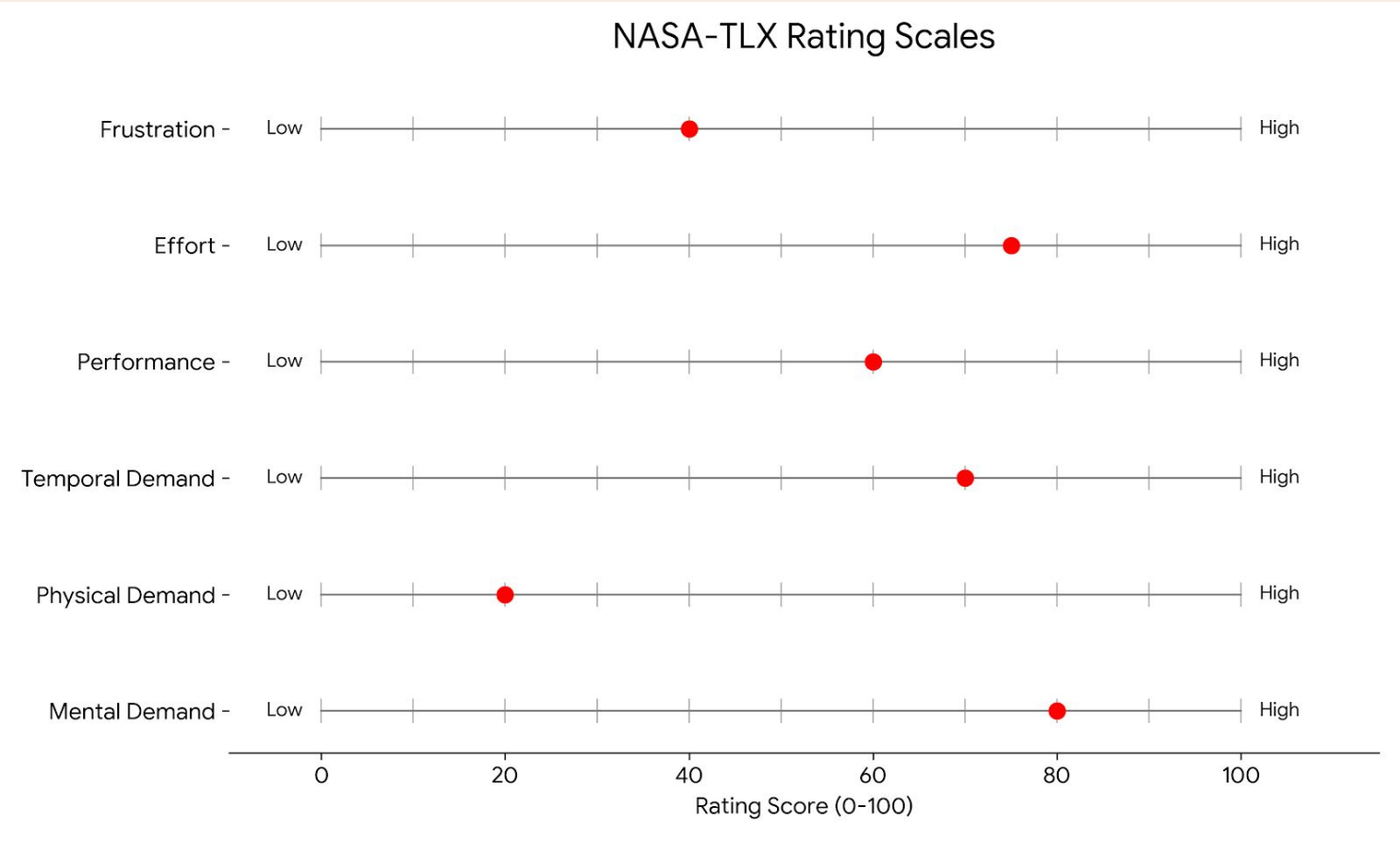
Research Objectives:

1. **Harmonize different eye-tracking datasets** into a fused feature space to resolve unit discrepancies.
2. **Justify the need for adaptation** by conducting a Domain Shift Analysis (via External Validation) and statistically quantify the performance degradation of static models when transferred to unseen data.
3. **Implement Cross-Domain Adaptation** to minimize the distance between the covariance matrices of the source and target domains.
4. **Develop a Test-Time Adaptation mechanism**, enabling the model to dynamically adapt to new users' physiological traits in real time.
5. **Integrate Explainability (XAI)** to validate the model's reliance on genuine physiological biomarkers rather than environmental artifacts

Preliminaries:

→ NASA Task Load Index (NASA-TLX)

- Subjective workload scores collected post-task
- Used to derive discrete cognitive load classes
- Enables supervised training across heterogeneous tasks

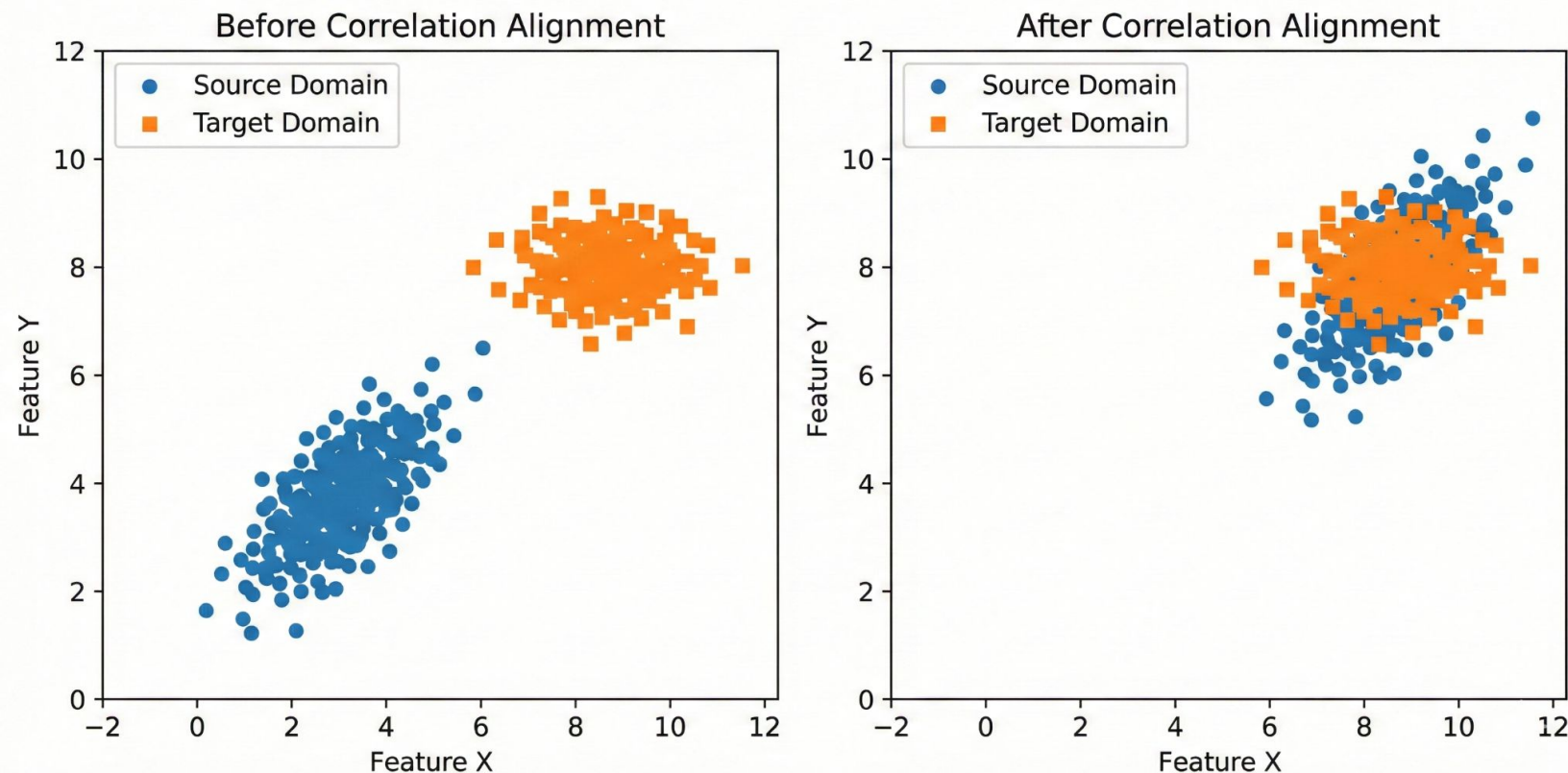


→ Correlation Alignment (CORAL)

- Aligns second-order feature statistics across datasets
- To reduce distribution mismatch between COLET & ADATABASE
- Model-agnostic and computationally lightweight

$$\begin{aligned}
 C_{source} &\leftarrow \text{Covariance}(X_{source}) + \lambda I \\
 C_{target} &\leftarrow \text{Covariance}(X_{target}) + \lambda I \\
 C_{align} &\leftarrow C_{source}^{-\frac{1}{2}} \cdot C_{target}^{\frac{1}{2}} \\
 X_{source}^{aligned} &\leftarrow X_{source} \cdot W_{align} \\
 Model_{adapted} &\leftarrow \text{TrainHistGradientBoosting}(X_{source}^{aligned}, \mathcal{F}_{top_k})
 \end{aligned}$$

Correlation Alignment

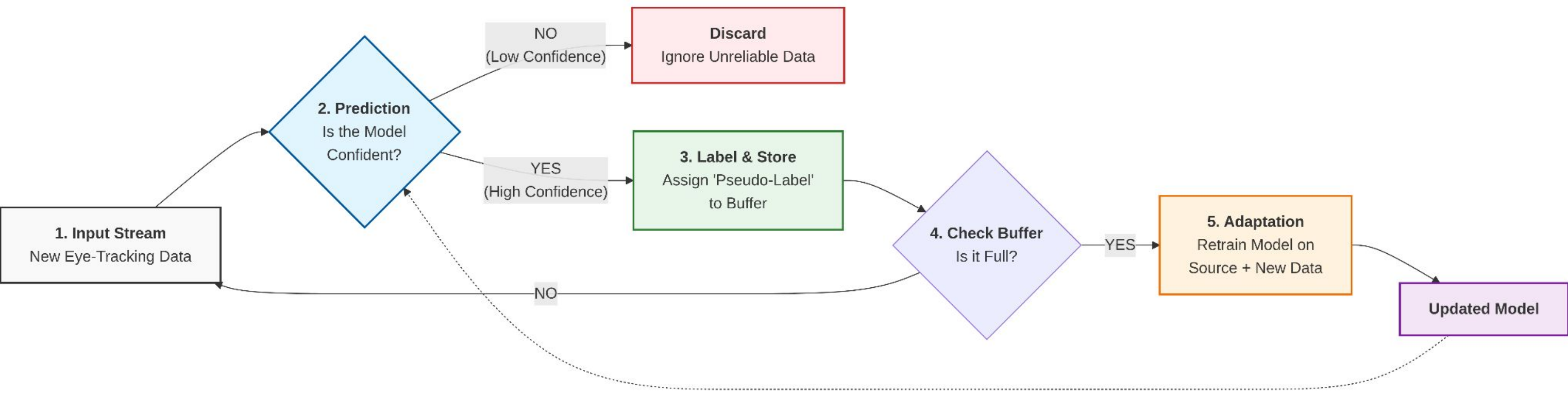


→ **Test-Time Adaptation (TTA)**

- Adapts the trained model during inference using unlabeled target-domain data
- Addresses distribution shift caused by inter-subject variability and task differences
- Model parameters are updated incrementally based on target data statistics
- Enables calibration-free deployment in real-world settings

→ **Iterative Pseudo-Labeling**

- Generates pseudo-labels from high-confidence model predictions
- Uses pseudo-labeled target samples to self-train the model
- Improves robustness to temporal drift and unseen users



Implementation:



Baseline Models:

A `HistGradientBoostingClassifier` was trained on COLET and tested on ADABase & vice versa to analyze the **Domain Shift**.

`learning_rate = 0.05, max_iter = 200, max_depth = 5, random_state = 42`

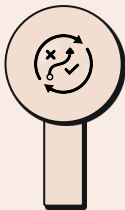
A **ResNet-MLP** architecture was used as another baseline comparison, to justify the ML-based pipeline over **Deep Learning**.

Dropout, $p = 0.3$; with **Batch Normalization**



Domain Adaptation (CORAL):

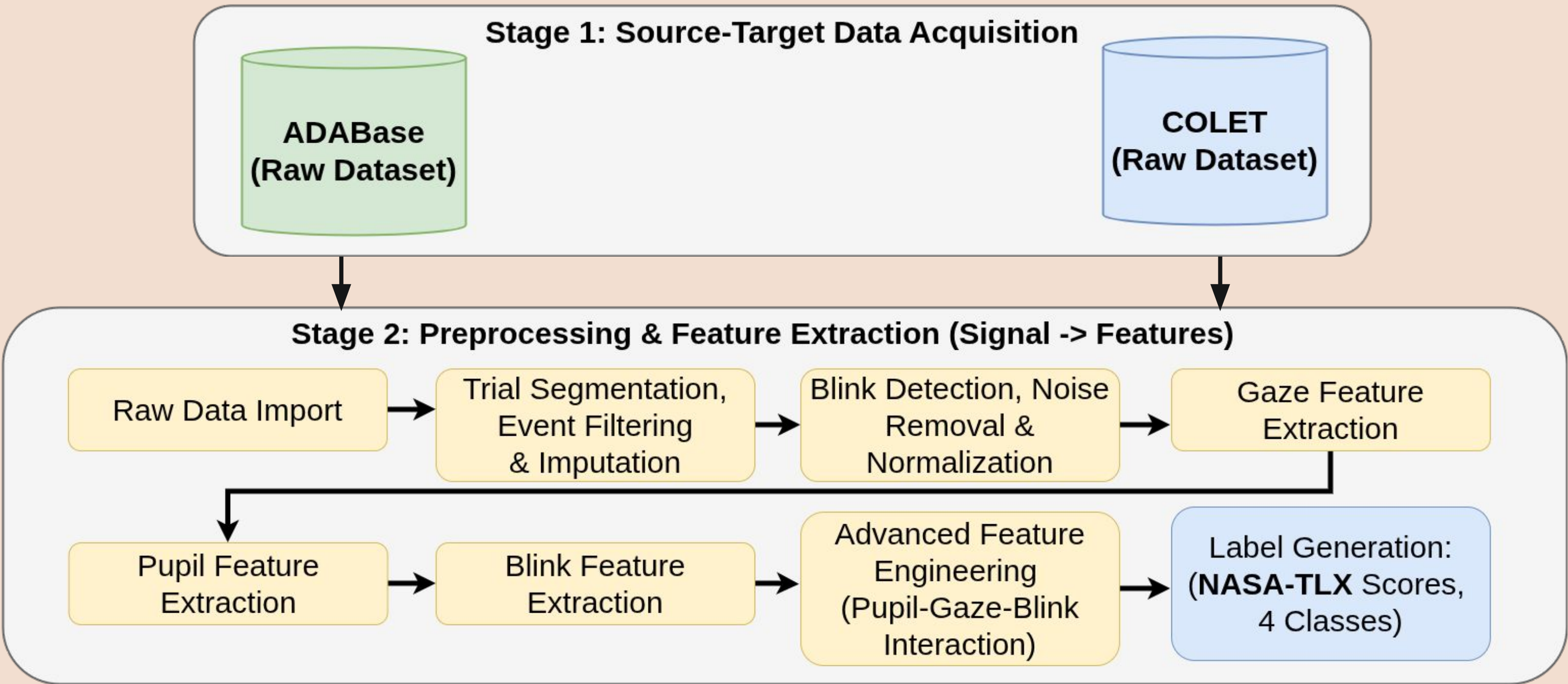
To bridge this shift, we aligned source covariance (C_{Source}) to target covariance (C_{Target}) to create an aligned feature space ($X_{Aligned\ Source}$).



Test-Time Adaptation (Iterative Pseudo-Labeling):

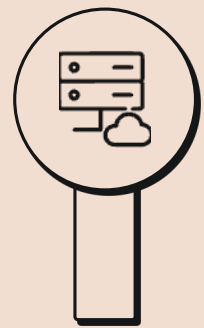
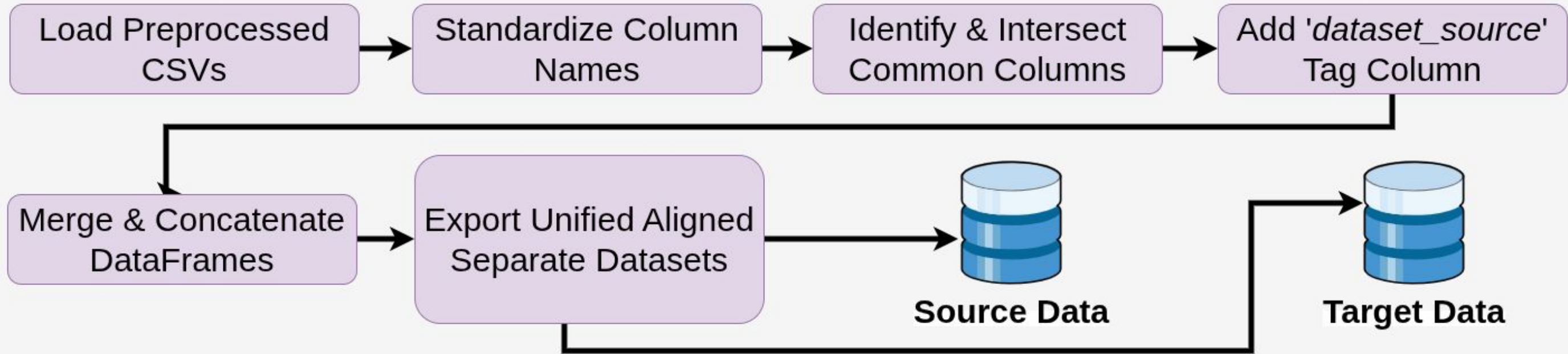
- Predicted probabilities on the live target stream.
- Filtered high-confidence predictions ($Conf_{max} > T_{high_confidence}$) into a pseudo-label buffer.
- Retrained the model dynamically on the original source + pseudo-labeled target data.

Methodology Pipeline:



Methodology Pipeline (Continued):

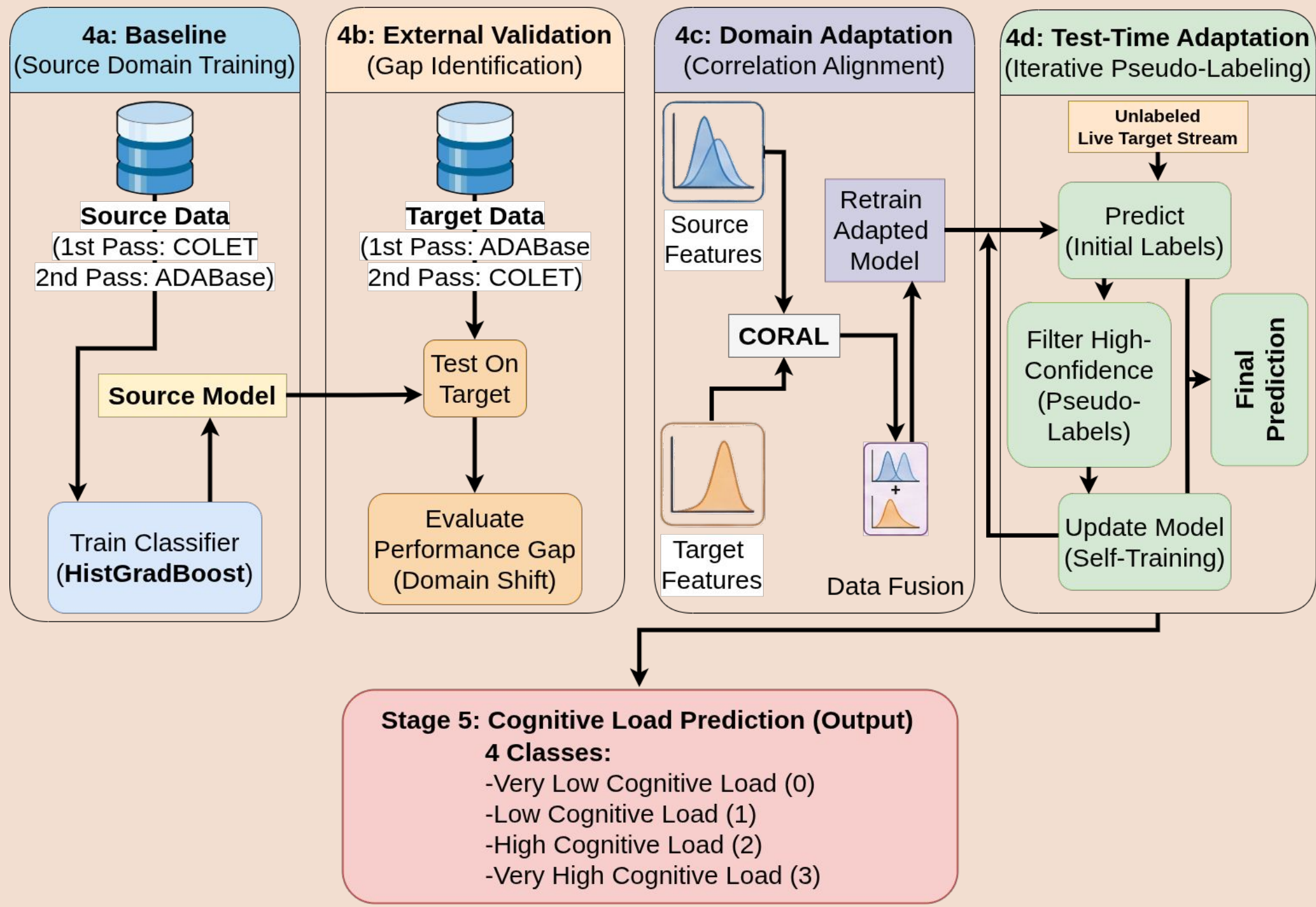
Stage 3: Data Alignment & Integration



Integration of COLET & ADABase datasets:

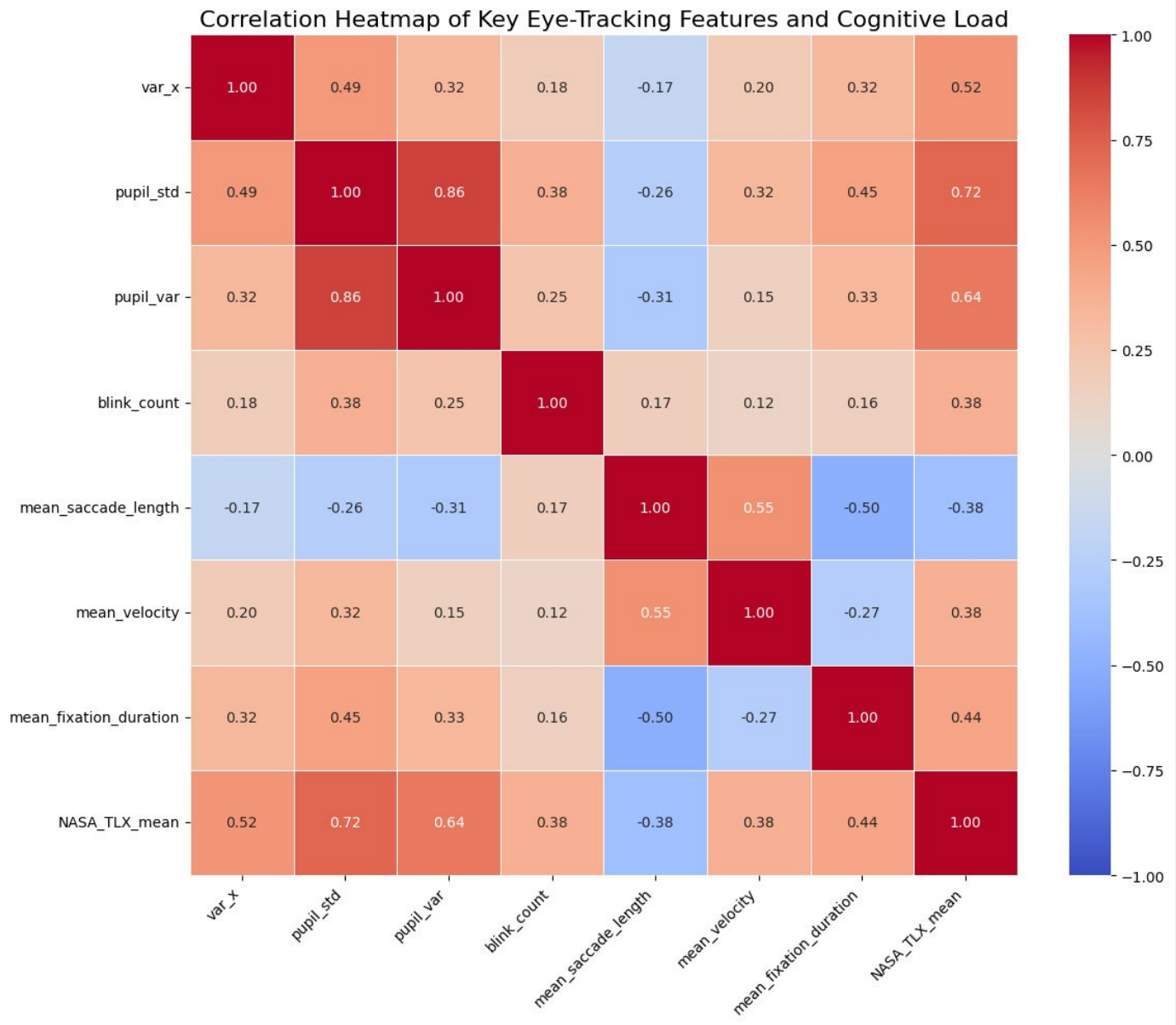
Extracted **gaze**, **pupil** and **blink** features, followed by rigorous **noise removal** and **signal normalization**.

Methodology Pipeline (Continued):



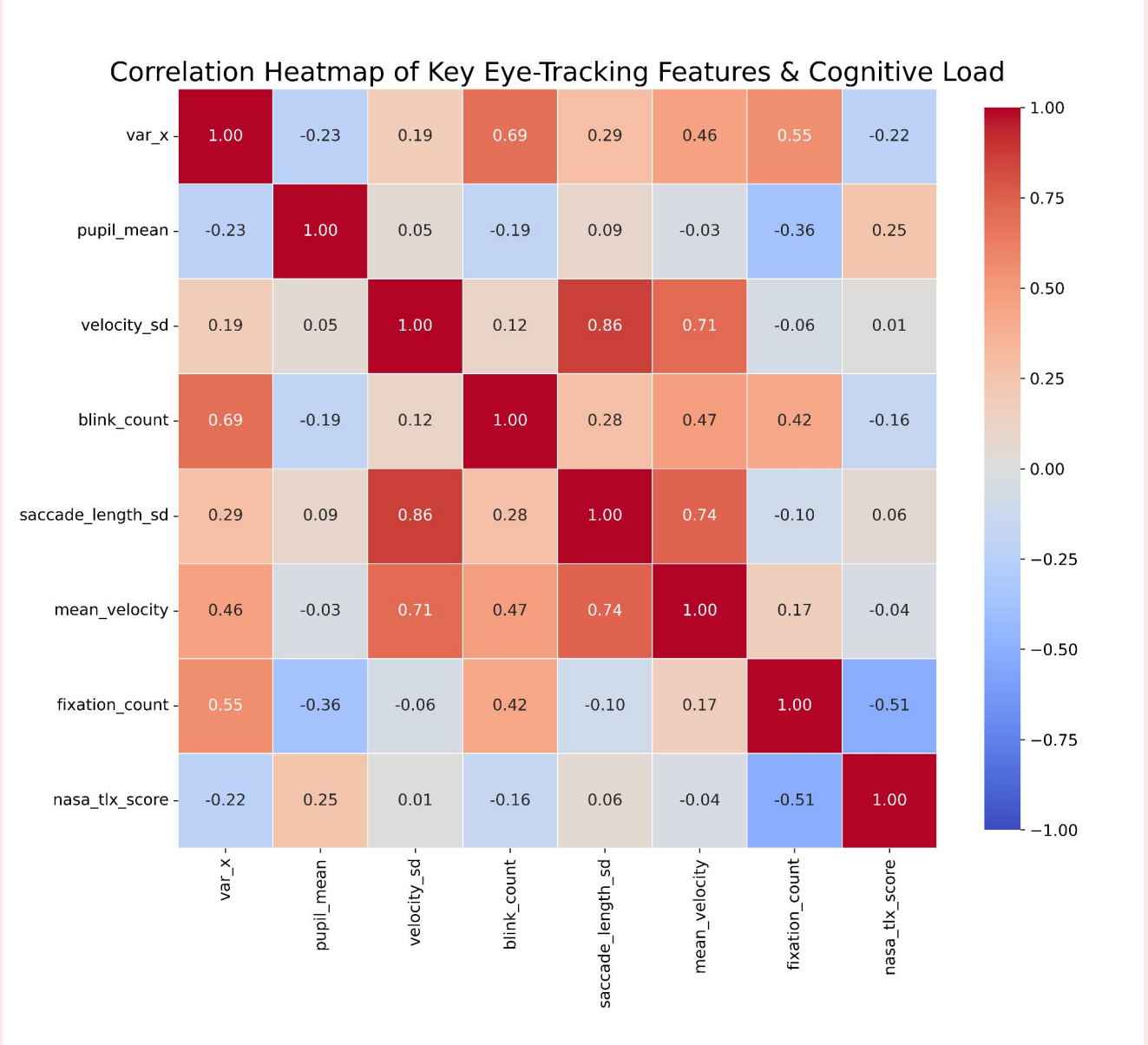
Exploratory Data Analysis (EDA):

The COLET dataset contains **47 subjects** performing **4 distinct visual tasks** which resulted in **188 unique observations**.



COLET Heatmap showing positive correlation between **pupil standard deviation** and **NASA-TLX score**, academically backed by **TEPR** (Beatty, 1982).

The ADABase dataset contains **30 subjects** performing **12 distinct driving tasks** with continuous signal streams which resulted in **360 unique observations**.



ADABase Heatmap showing positive correlation between **pupil dilation mean** and **NASA-TLX score**.

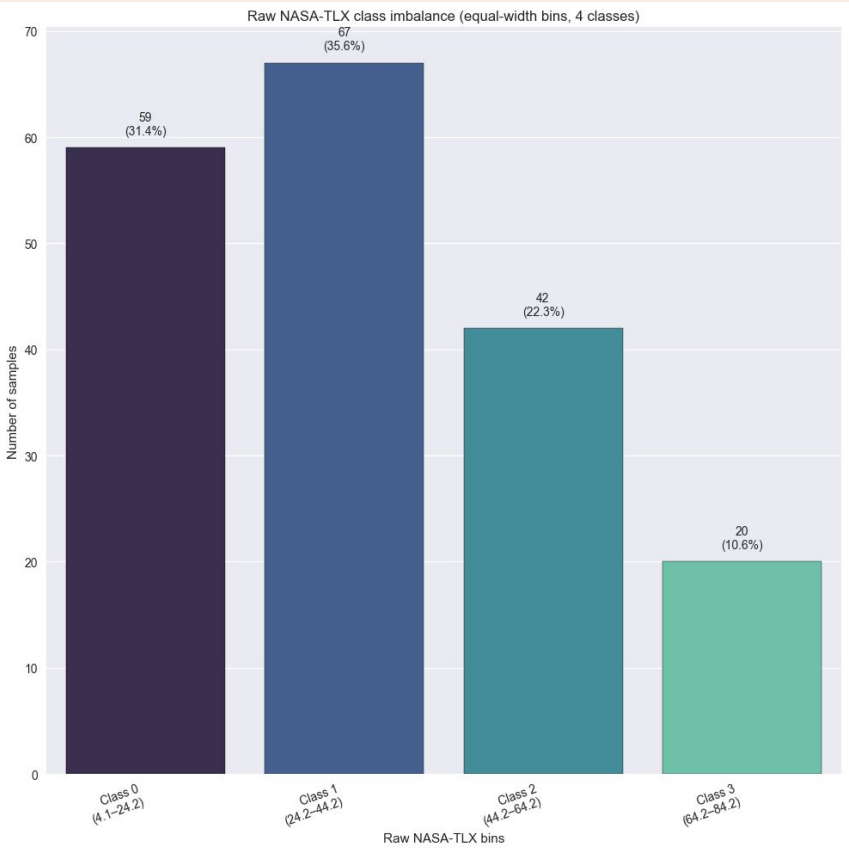
Data Cleaning & Transformation:

Feature Extract: Extracted 33 basic metrics and expanded to **94 advanced engineered features**.

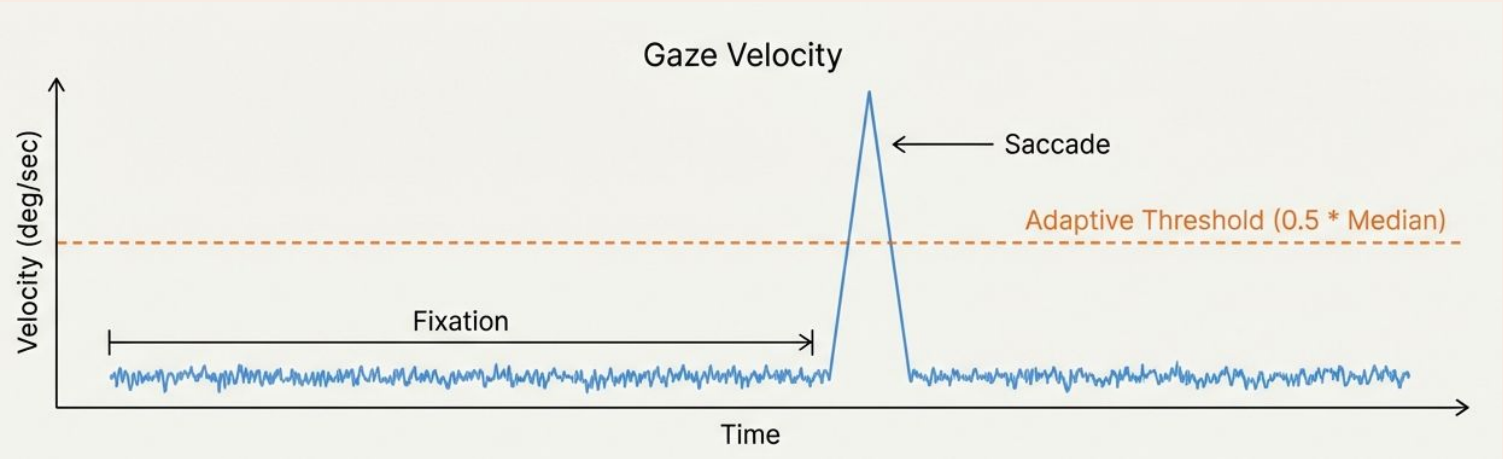
Imputation: Applied **Median and Zero Imputation** to avoid data leakage.

Composite Indices: Cognitive states were proxied by new feature combinations like **Pupil Diameter/Saccade Velocity**, also known as Index of Cognitive Activity (Marshall, 2002).

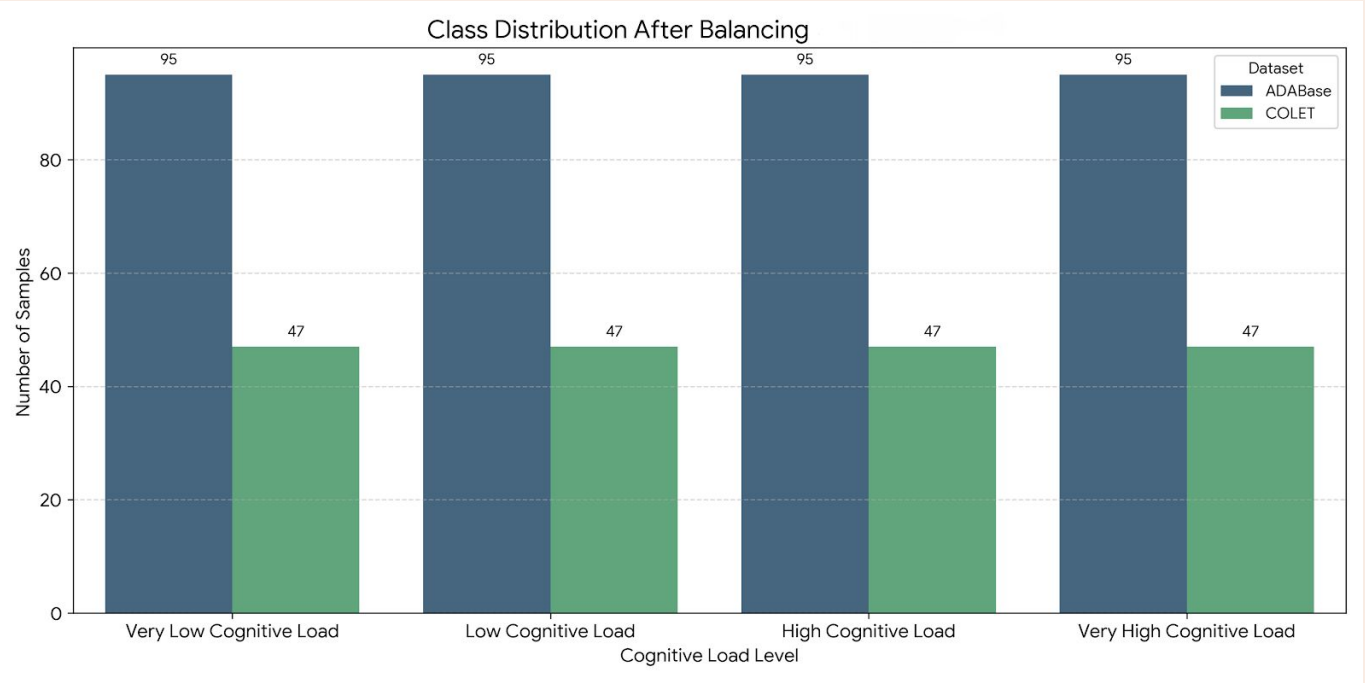
Interaction Features: Pupil diameter multiplied by gaze coordinates to deal with **gaze-dependent pupil responses** and remove cosine error (Hayes & Petrov, 2016).



Class Imbalance with equal-width bins.



Fixation and Saccade I-VT Algorithm (Salvucci & Goldberg, 2000).



Balanced Classes Obtained.

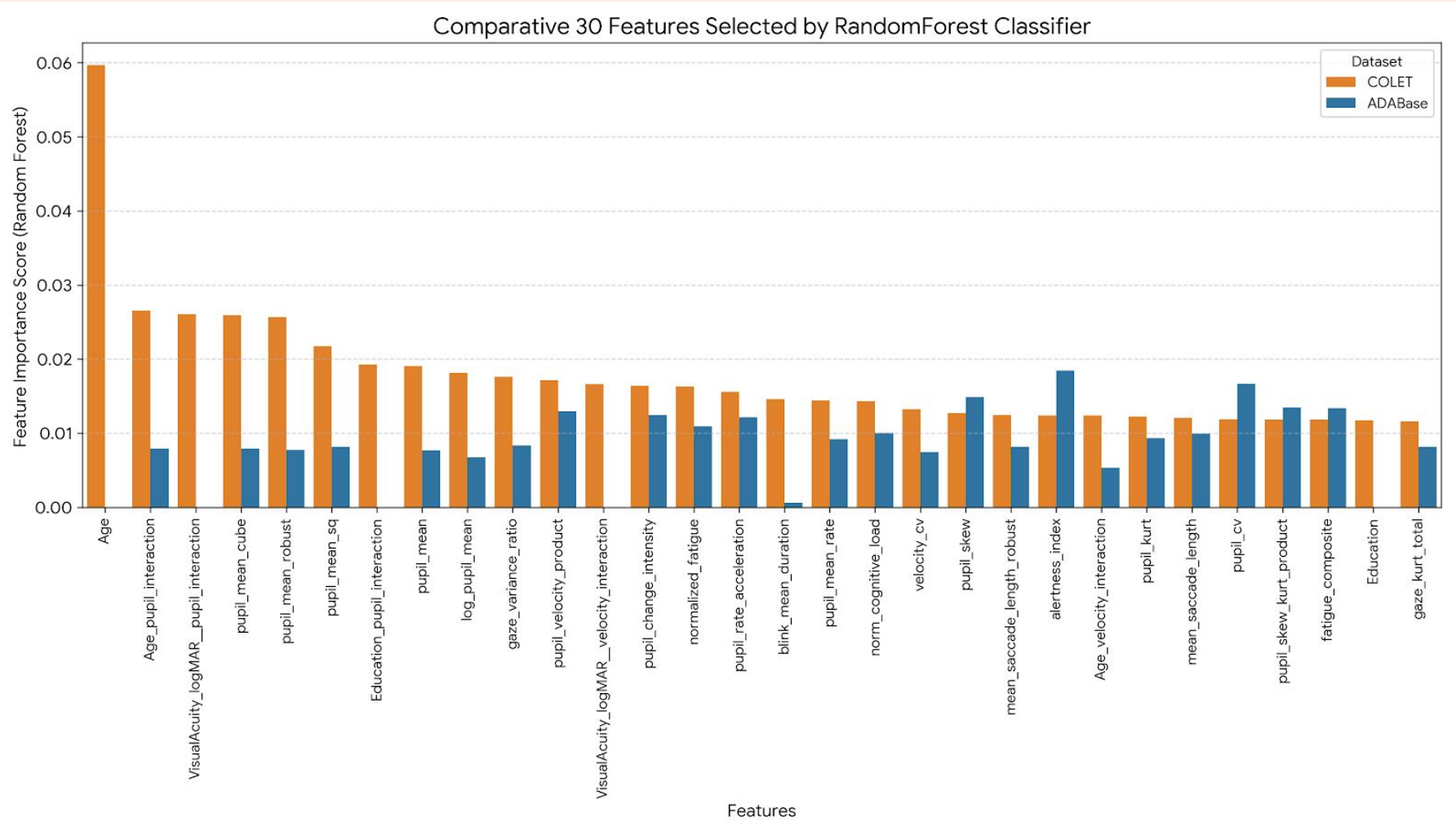
Class Imbalance Handling:

Quantile Binning for COLET: Partitioned the NASA-TLX scores into 4 equal parts, i.e., from Very Low to Very High.

Oversampling for ADABase: Applied Random Oversampling to match the number of instances in the minority class with the majority class.

Data Integration:

Following the independent preprocessing of the source domain (COLET) and the target domain (ADABase) individually, a stringent data alignment phase is performed, and finally getting aligned feature spaces despite disparate raw formats.



pupil_mean_cube	alertness_index	...	gaze_variance_ratio	nasa_tlx_mean
12.21	0.45	...	2.10	16.7
10.55	0.52	...	1.85	9.2
15.30	0.33	...	3.44	45.0
14.12	0.38	...	2.95	44.2
9.88	0.61	...	1.55	23.3

Table 4.2: Excerpt of COLET trials showing key features

pupil_mean_cube	alertness_index	...	gaze_variance_ratio	nasa_tlx_mean
19.77	0.08	...	1.07	7.00
20.28	0.14	...	1.06	11.33
20.85	0.15	...	0.99	12.00
18.54	0.18	...	0.37	1.50
20.98	0.10	...	0.59	4.83

Table 4.3: Excerpt of ADABase trials showing key features

Feature Intersection:

Filtered dataset-specific artifacts to include only **94 shared physiological features**.

Source Tagging:

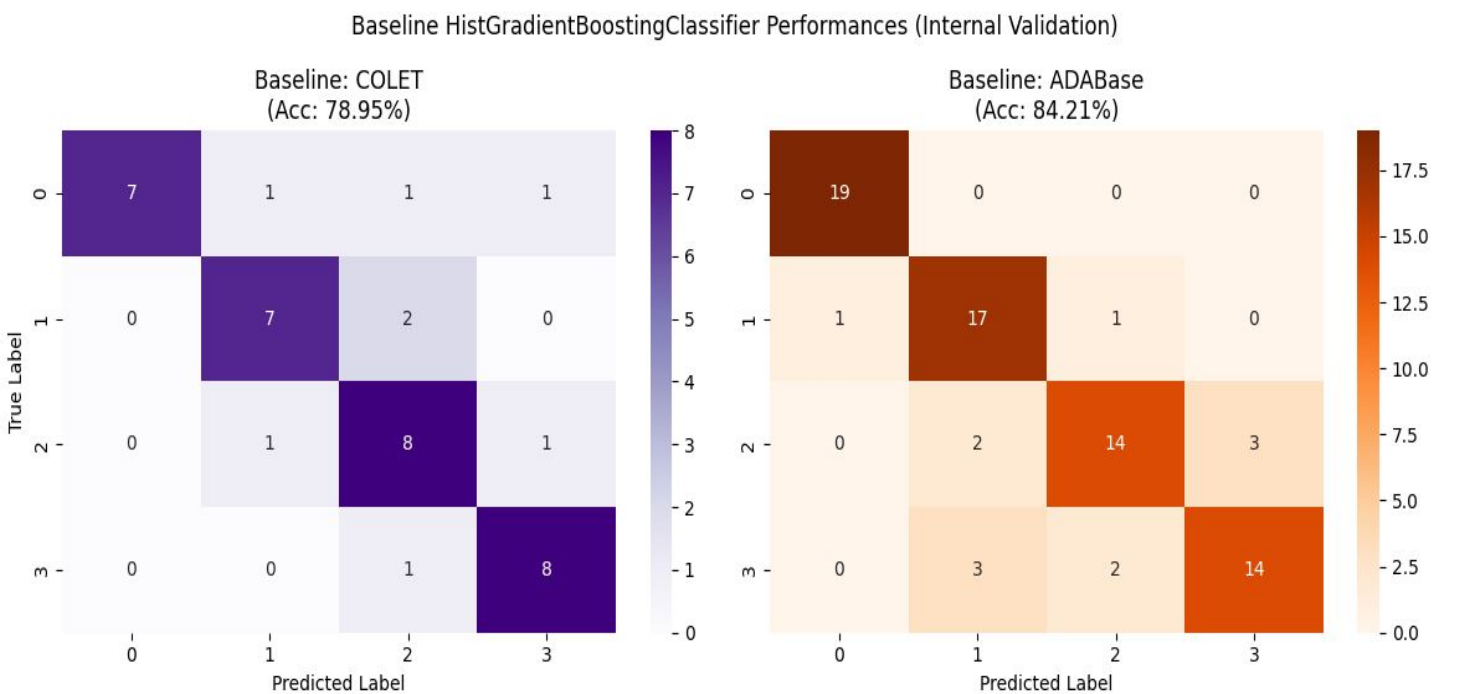
Added a metadata tag "dataset_source" to all trials for domain identification.

Baseline Performance Evaluation:

- Evaluating the baseline `HistGradientBoostingClassifier` (Internal Validation).
- First Pass: Trained on `COLET`.
- Second Pass: Trained on `ADABase`.
- Decision Tree-based *ensemble* models are adequate for the Tabular Datasets.
- Internal Validation scores are the *Theoretical Upper Limit* for the performance comparisons.

COLET Accuracy = 78.95%	COLET Precision:	COLET Recall:	COLET F1 Score:
Macro Average:	0.8111	0.7917	0.7927
Weighted Average:	0.8123	0.7895	0.7918

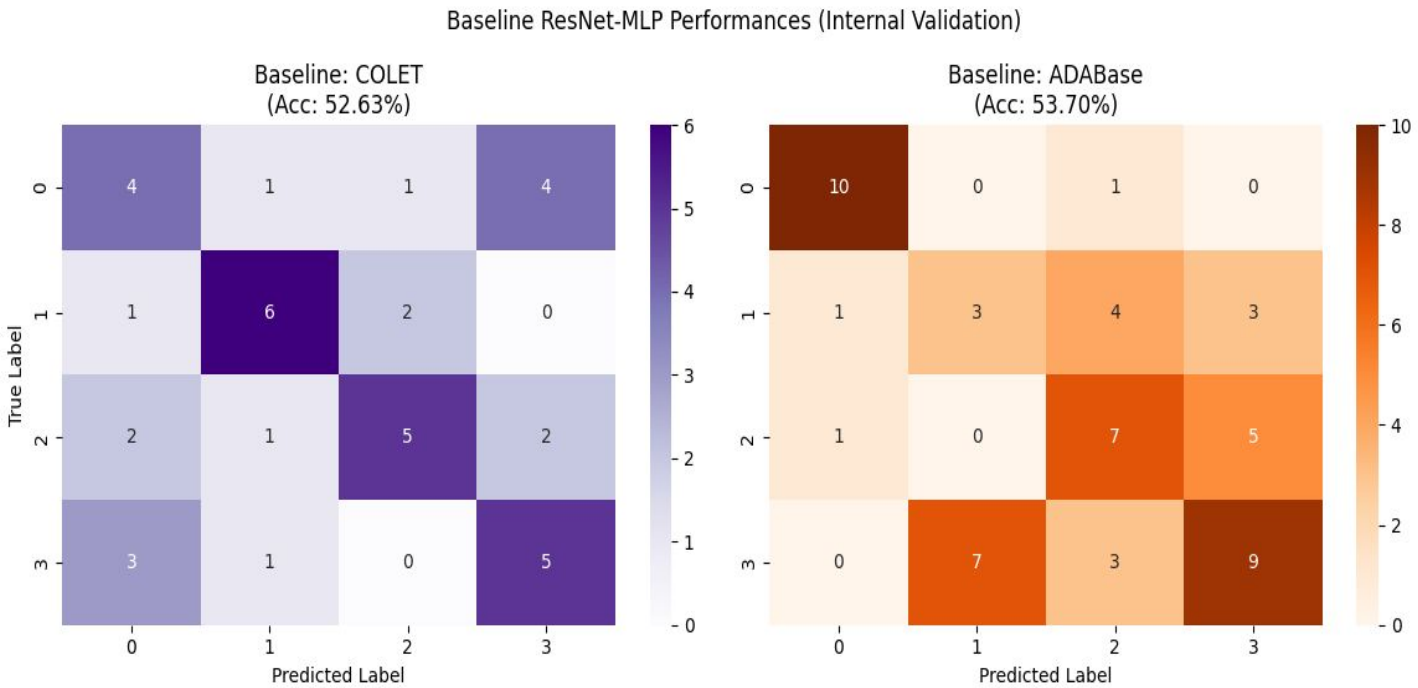
ADABase Accuracy = 84.21%	ADABase Precision:	ADABase Recall:	ADABase F1 Score:
Macro Average:	0.8424	0.8421	0.8398
Weighted Average:	0.8424	0.8421	0.8398



Why not Deep Learning?

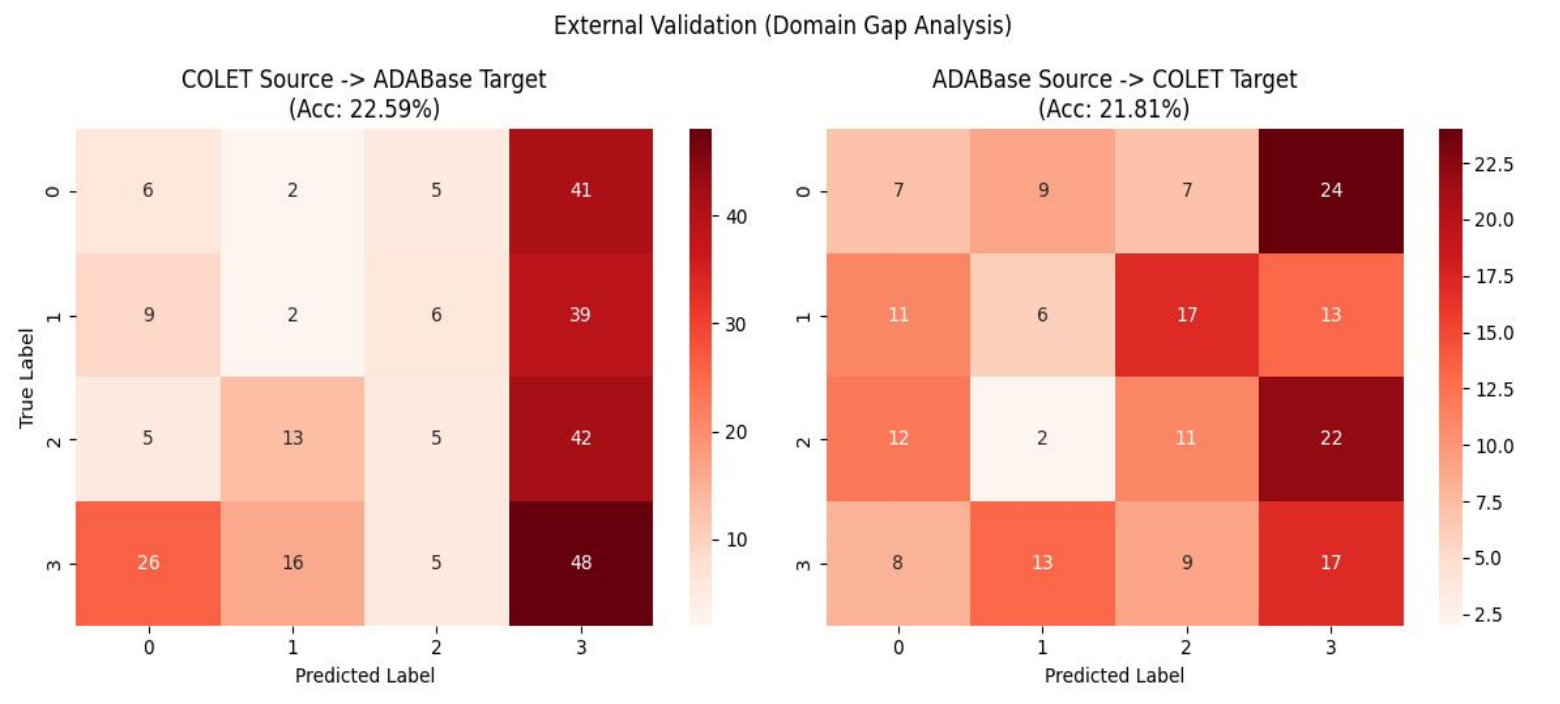
- Evaluating the baseline **ResNet-MLP** architecture (Internal Validation).
- First Pass: Trained on **COLET**.
- Second Pass: Trained on **ADABase**.
- Limited observations on **Tabular Datasets** lead to *underfitting* on DL-based architectures.
- DL implementations work better on the likes of **Time Series EEG** data (Grinsztajn et al., 2022).

ResNet COLET Accuracy = 52.63%	ResNet COLET Precision:	ResNet COLET Recall:	ResNet COLET F1 Score:
Macro Average:	0.5366	0.5306	0.5306
Weighted Average:	0.5353	0.5263	0.5278
ResNet ADABase Accuracy = 53.70%	ResNet ADABase Precision:	ResNet ADABase Recall:	ResNet ADABase F1 Score:
Macro Average:	0.5324	0.5485	0.5388
Weighted Average:	0.5295	0.5370	0.5316



Domain Shift Analysis:

- Analyzing the Performance Gap when Domain Shifts occur (External Validation).
- Using the baseline **HistGradientBoostingClassifier**.
- First Pass: Source **COLET** -> Target **ADABase**
- Second Pass: Source **ADABase** -> Target **COLET**
- **56.36%** drop observed from the baseline on the first pass.
- **62.4%** drop observed from the baseline on the second pass.
- Accuracies dropped below *Random Guessing*.



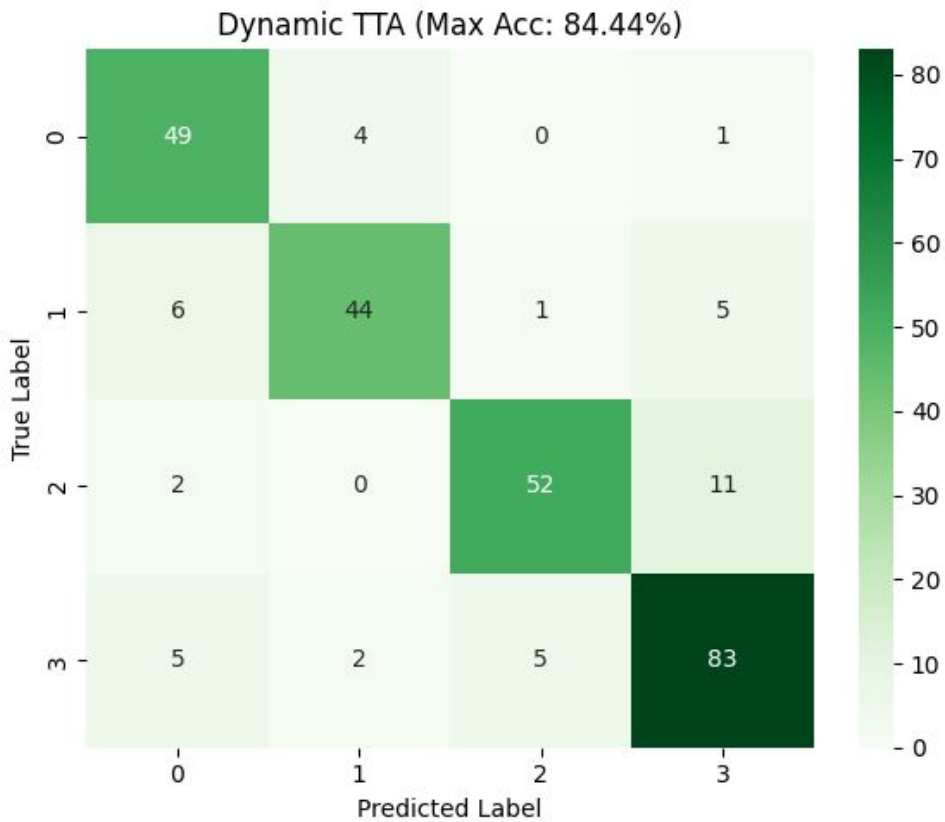
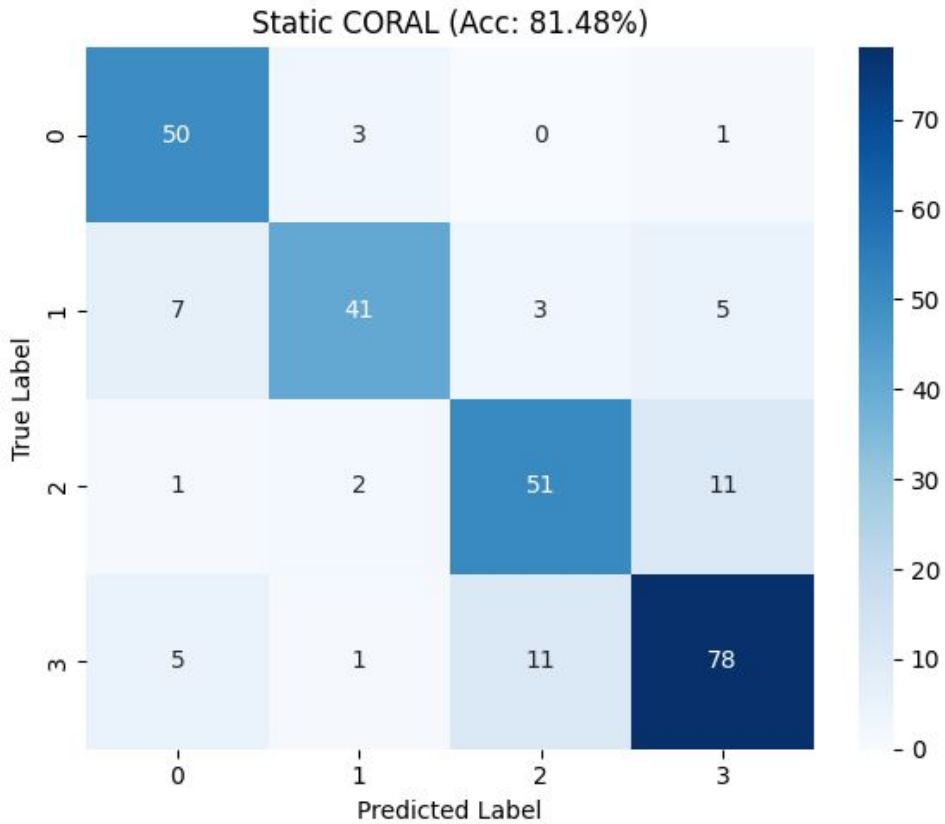
COLET to ADABase Accuracy = 22.59%	COLET to ADABase Precision:	COLET to ADABase Recall:	COLET to ADABase F1 Score:
Macro Average:	0.1779	0.1823	0.1609
Weighted Average:	0.1953	0.2259	0.1888

ADABase to COLET Accuracy = 21.81%	ADABase to COLET Precision:	ADABase to COLET Recall:	ADABase to COLET F1 Score:
Macro Average:	0.2145	0.2181	0.2097
Weighted Average:	0.2145	0.2181	0.2097

Cross-Domain & Test-Time Adaptation:

- Applying **Correlation Alignment** to the **HistGradientBoostingClassifier**.
- Source **COLET**; Target **ADABase**.
- Making the model *Self-Supervised* via **Iterative Pseudo-Labeling**.
- Balanced iteration count = **10**, Confidence Threshold = **0.85**.

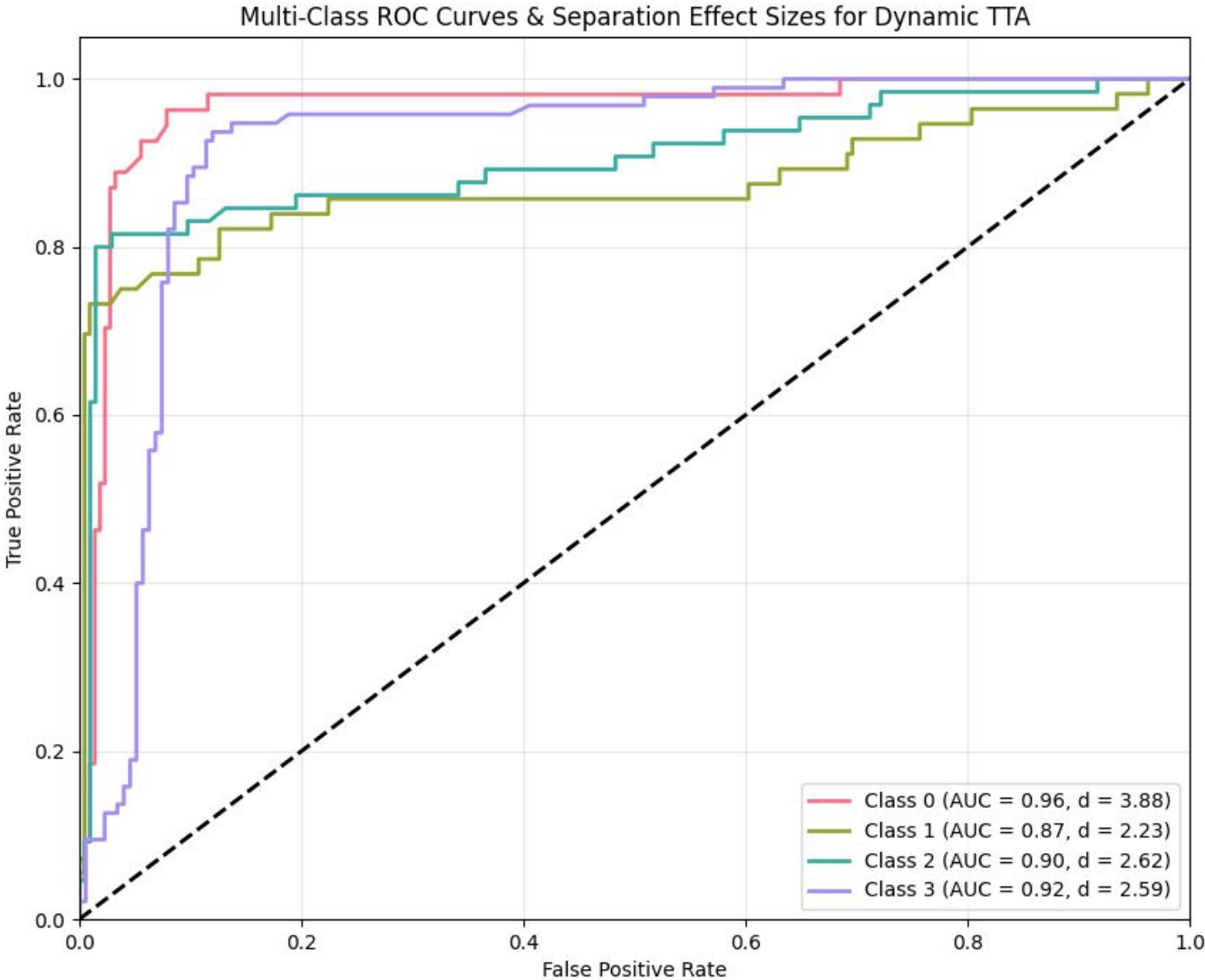
Static CORAL Accuracy = 81.48%	Static CORAL Precision:	Static CORAL Recall:	Static CORAL F1 Score:
Macro Average:	0.8179	0.8159	0.8141
Weighted Average:	0.8174	0.8148	0.8138
Dynamic TTA Accuracy = 84.44%	Dynamic TTA Precision:	Dynamic TTA Recall:	Dynamic TTA F1 Score:
Macro Average:	0.8492	0.8417	0.8430
Weighted Average:	0.8485	0.8444	0.8442



Statistical Analysis:

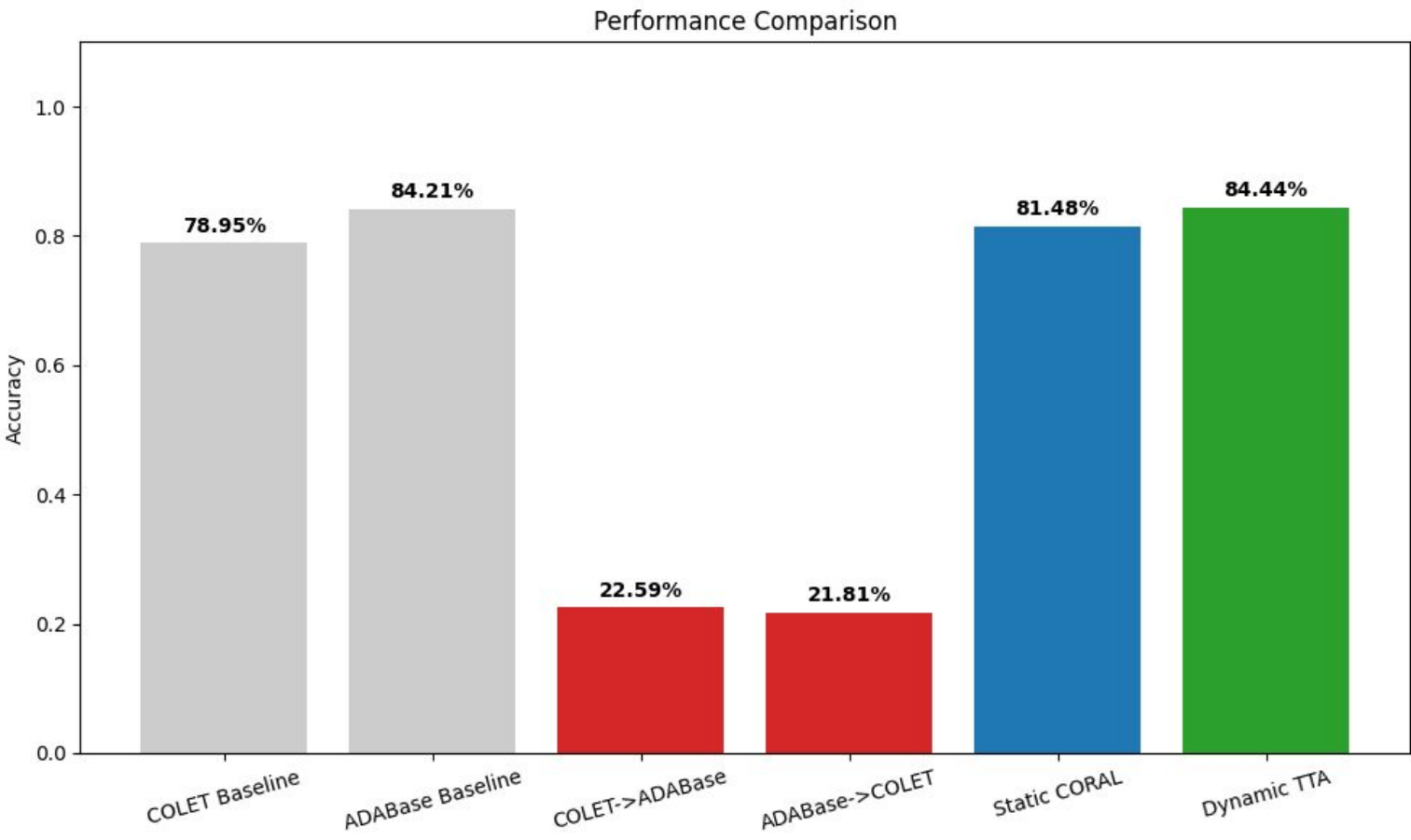
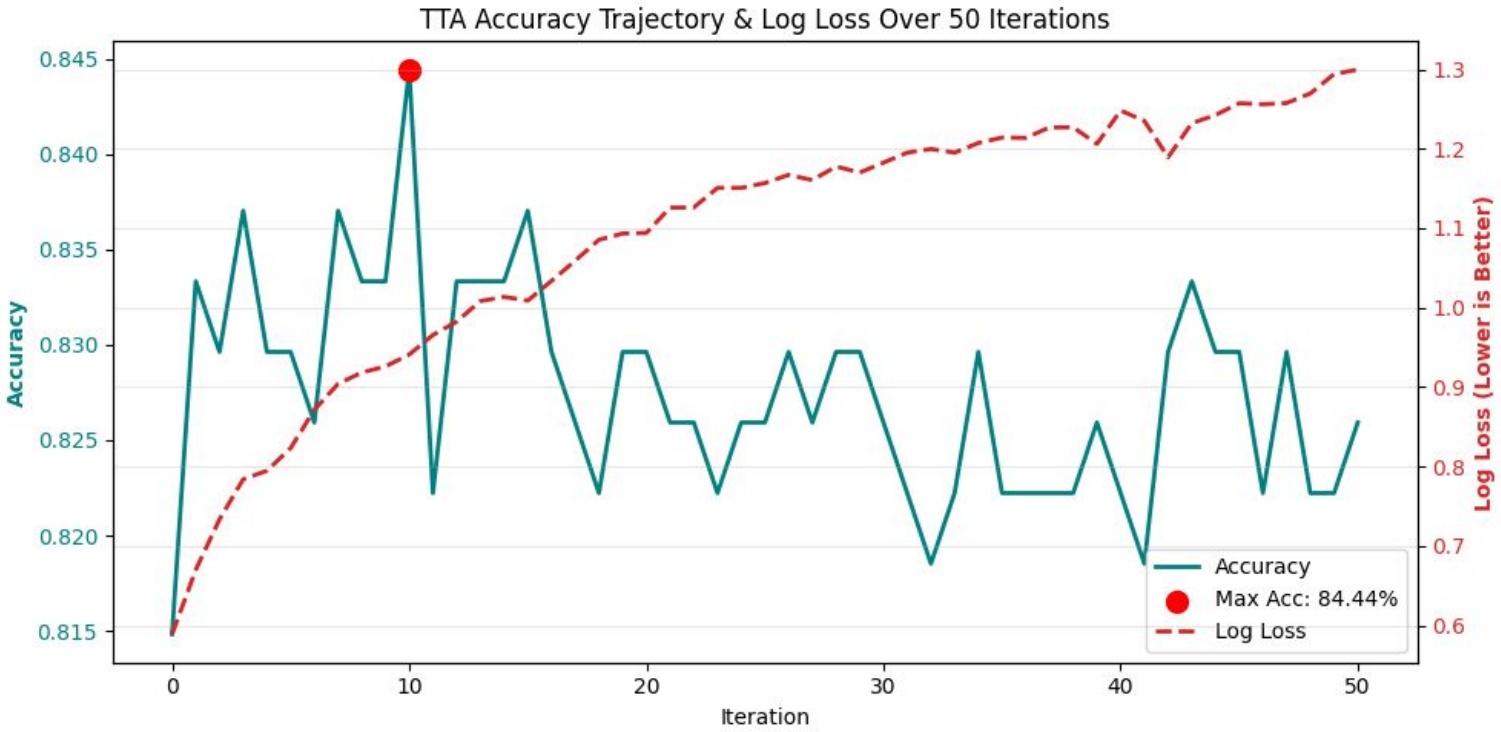
- Evaluated on a Ryzen 7 7840HS + RTX4060 + 16GB LPDDR5x 6400MHz (CPU-only mode, with *CUDA disabled*).
- Multiclass ROC shows **high AUC** and *non-overlapping* curves.
- Cohen’s Effect Size coefficients, $d > 0.8$ denotes a Large effect; $d > 3$ denotes *nearly complete separation*.
- The model can effectively distinguish between:
 - C0 & C1; C0 & C2; C0 & C3
 - C1 & C2; C1 & C3
 - C2 & C3
- Difficulties in pre-TTA models with separating between Classes C2 & C3 addressed.

	Accuracy	Log Loss	Batch Time (s)	Cumulative Time (s)
Iteration 10	0.8444	0.9409	0.5634	5.0578
Iteration 20	0.8296	1.0940	0.6928	11.3697
Iteration 30	0.8259	1.1825	0.6959	18.1747
Iteration 40	0.8222	1.2480	0.8140	25.7121
Iteration 50	0.8259	1.2996	0.8474	33.7822



Adjustments & Comparisons:

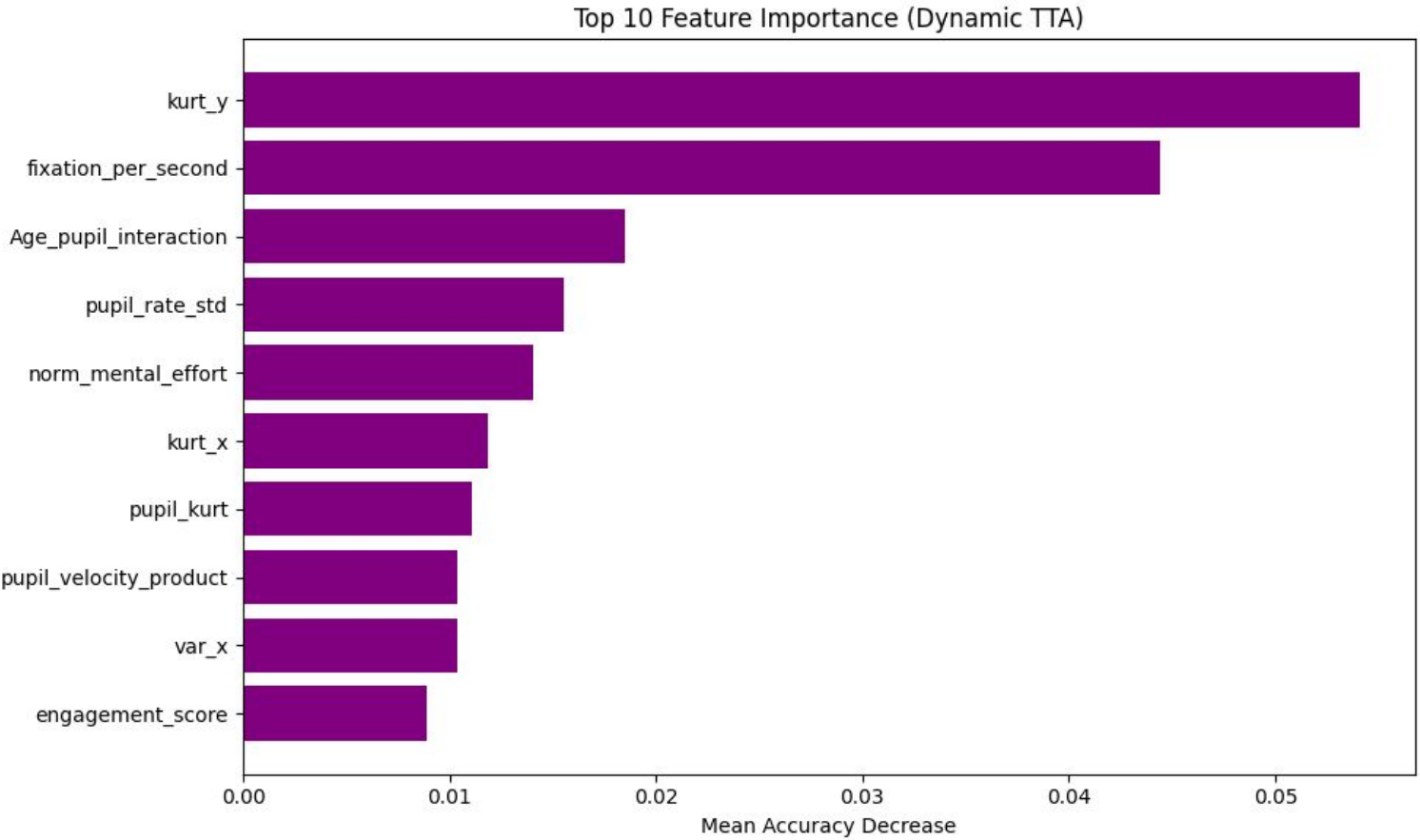
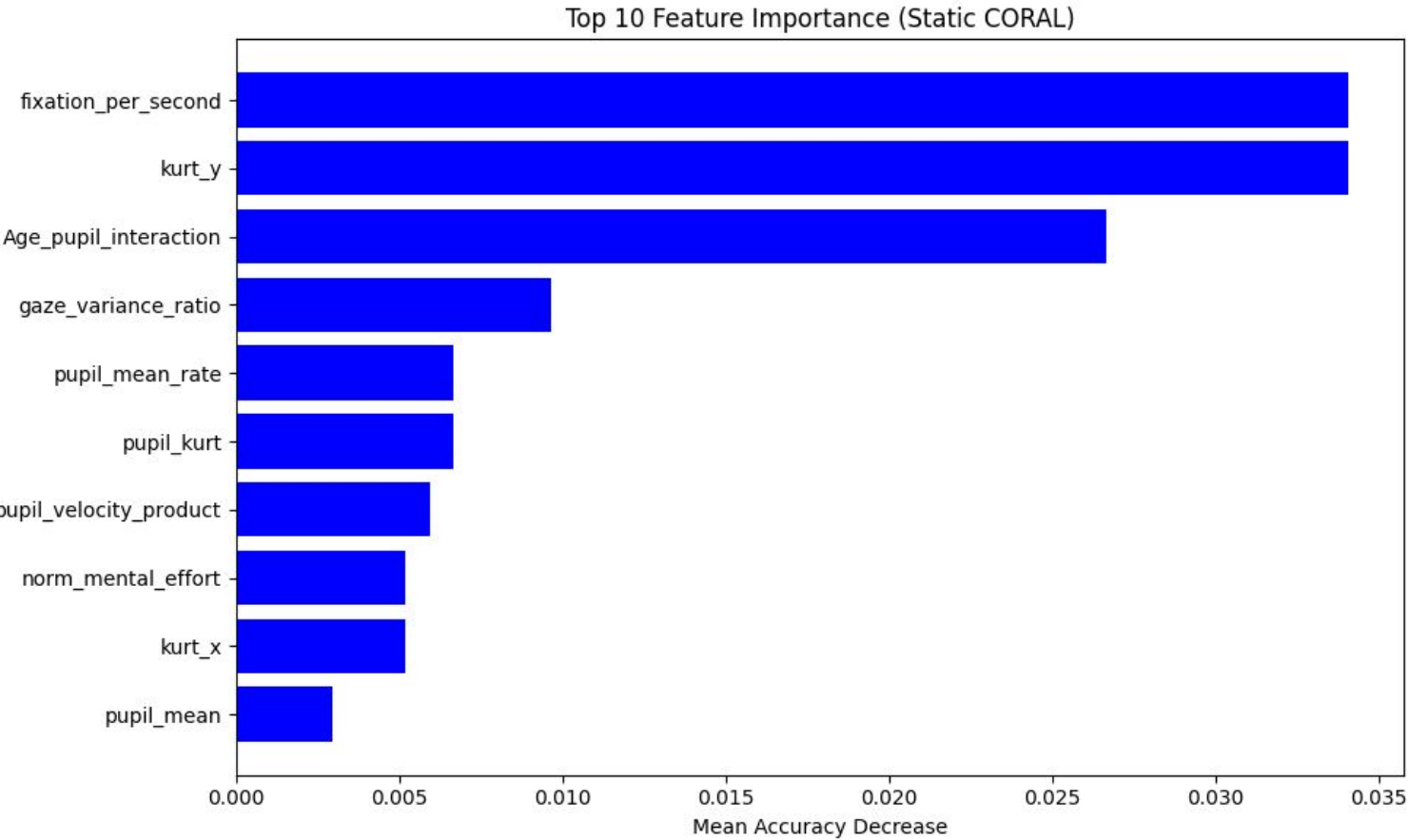
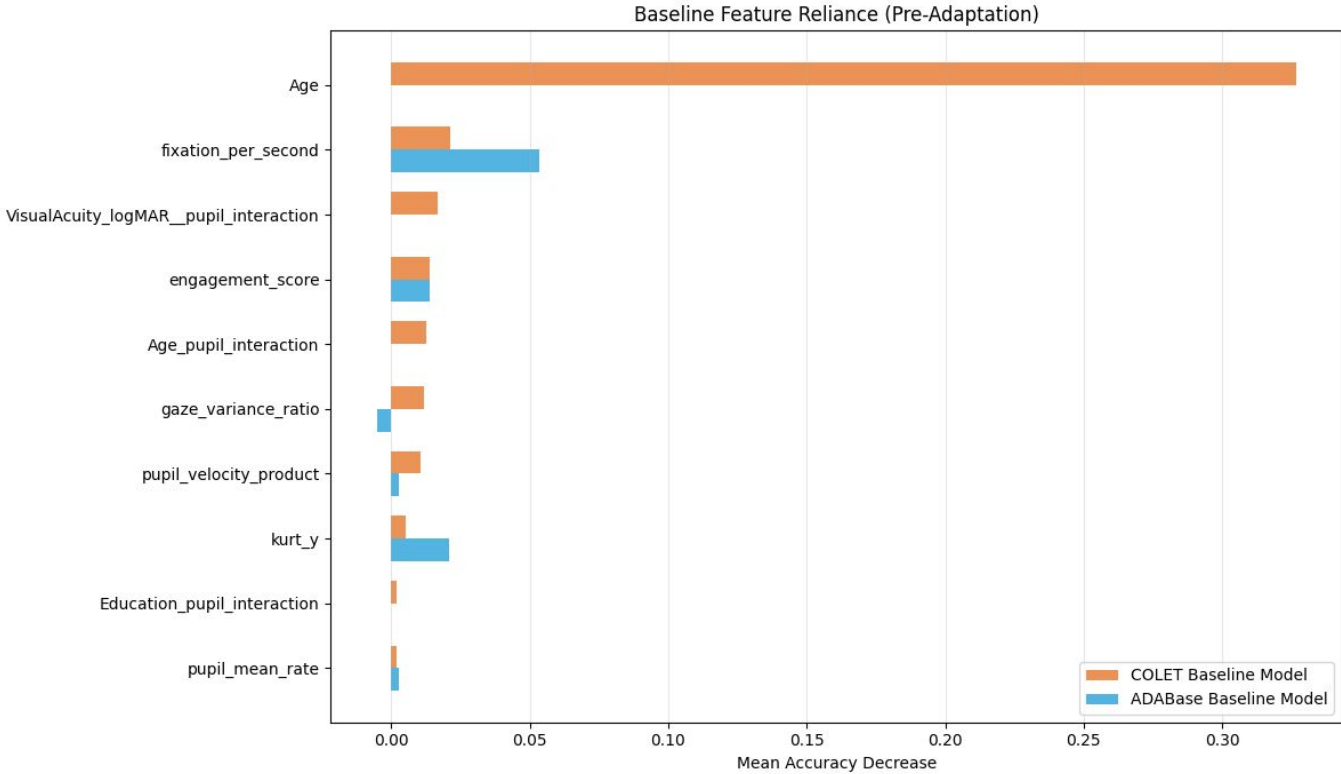
- Performance (**Accuracy**) vs. Cost (**Log Loss**) analyzed over **50 iterations**.



- Increasing **Log Loss** suggests that the model becomes *overconfident* if no early-stopping conditions are set.
- Below a **Confidence Threshold** of **0.85**, the model is too *lenient* and above it, the model is too *rigid* (Lee, 2013).

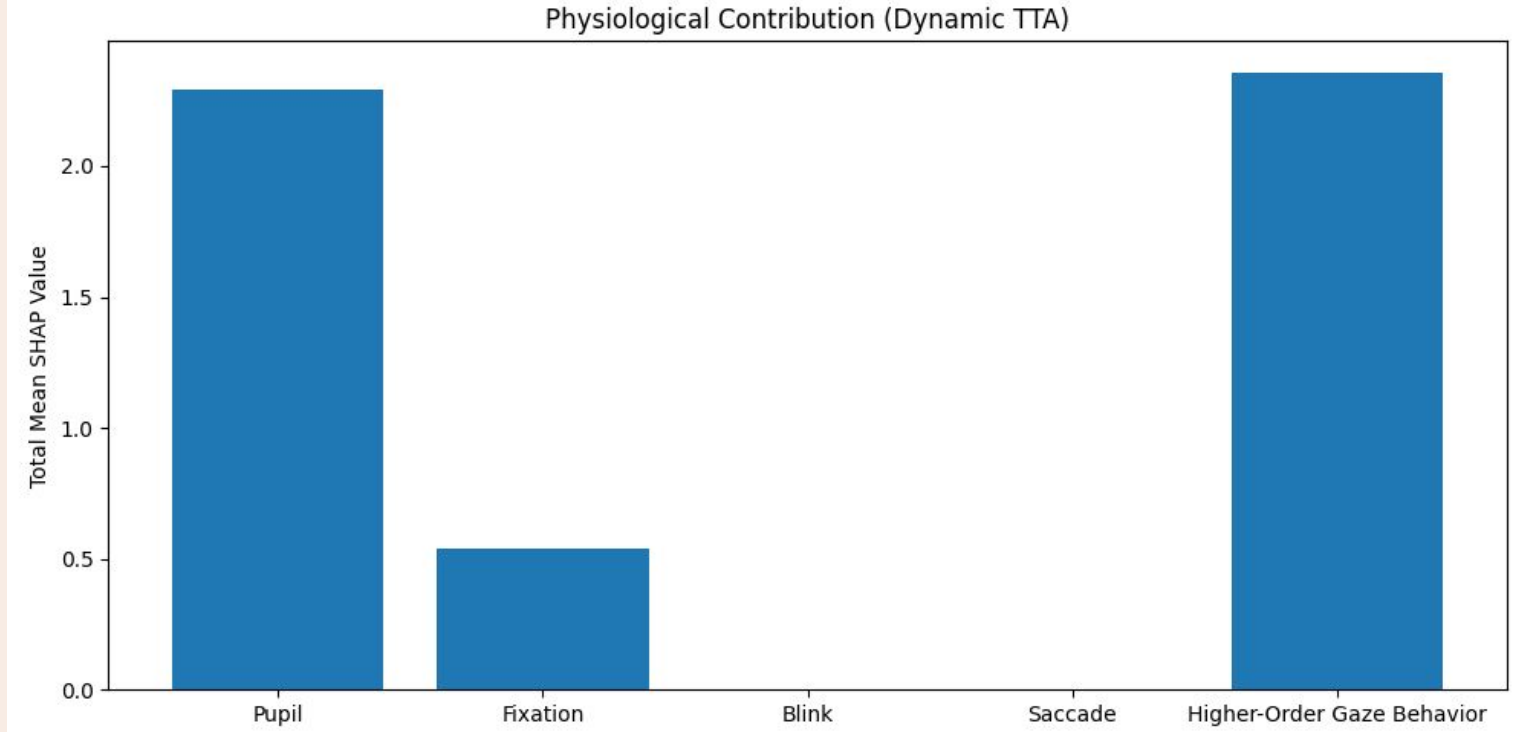
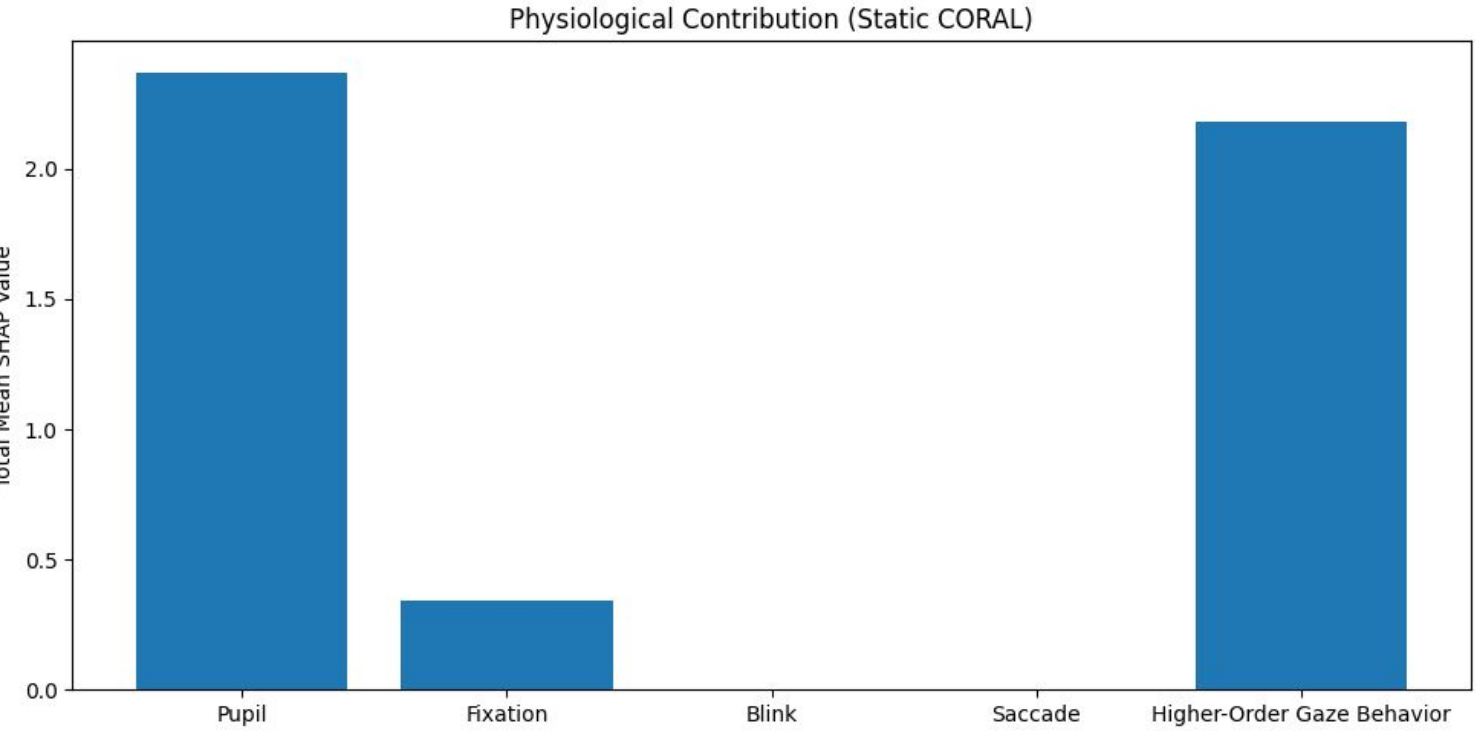
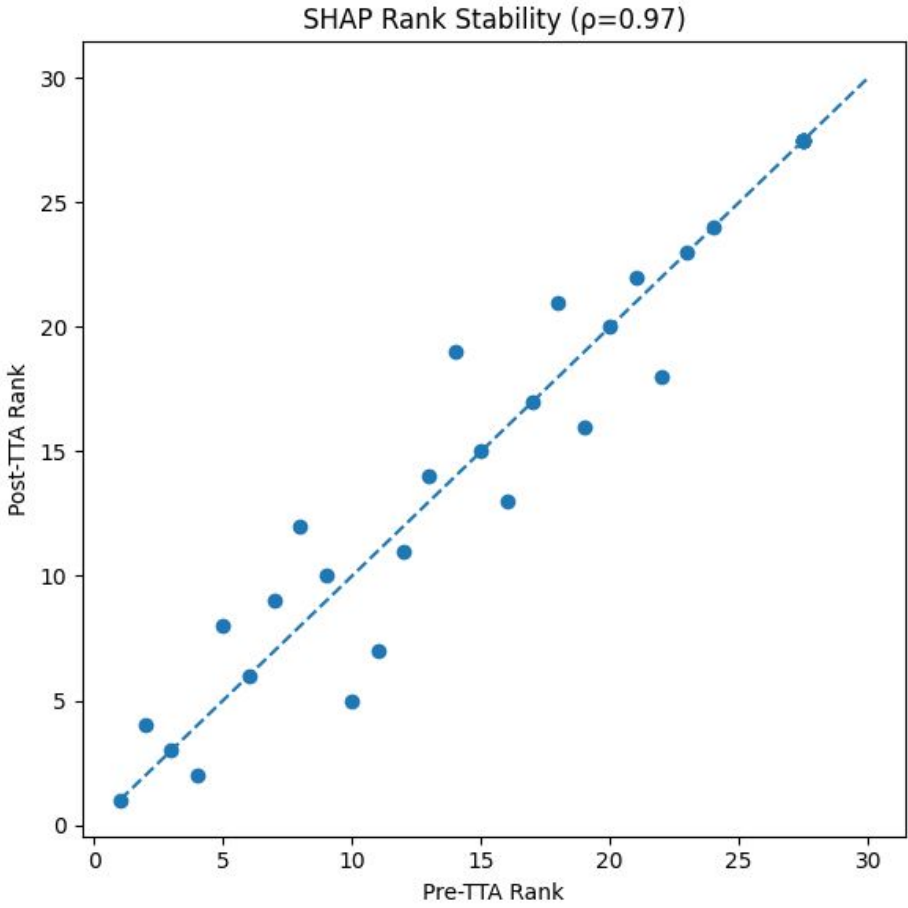
Interpretability of the Models:

- Analysis of **Global Feature Importance** using **Average Marginal Contribution (SHAP Values)** & **Local Feature Importance** via **Permutation Scores**.
- Discrepancy in feature importances of the baseline models (positive correlation in COLET, yet 0, or even **negative permutation importance** in ADABase), highlighting the **domain shift gap**.
- *Subjective* features prioritized **pre-adaptation**.
- *Domain-invariant* features correlated to the label - **kurtosis** of the gaze in y-axis, **fixation** per second & **pupil** metrics, prioritized **post-adaptation**.



Physiological Group Contributions & SHAP Rank Stability:

- $\rho > 0.75$ demonstrates strong **consistency**, so the learned representation of CL is stable *cross-domain*.
- **Pupil**-related features contribute the most in predicting CL *post-adaptation*, aligning with the findings of Pomplun & Sunkara (2019).
- **Blink** data in the source distribution was **sparse** and often *task dependent* and *subject-specific*.
- **Saccades** are **short-lived** & hard to capture, often recorded indirectly as a *fixation-derived metric*.
- **Behavioral** features, obtained from the feature engineering chain by calculating **interaction scores**, denoted as **higher-order gaze behavior**.



Conclusion:

Limitations:

- NASA-TLX scores are subjective measures of perceived task load.
- Live target data still requires undergoing the same preprocessing, feature engineering & feature alignment steps.
- 4-class classification limits the model's peak performance compared to a 3-class CL classification problem.
- As HistGradientBoostingClassifier is an ensemble model based on decision trees, the entire decision tree needs to be rebuilt each time the model updates itself.

Future Work:

- Integrating multimodal sensor fusion and comparing the effectiveness of time series EEG data against tabular eye-tracker data for CL estimation
- Implementing a Deep Learning-based pipeline for Cross-Domain & Test-Time Adaptation that works well on tabular data
- Developing a signal chain that undergoes minimal preprocessing to produce schema-agnostic predictions
- Creating a standardized format for sampling user gaze information, such that shifting targets do not require prior feature alignment
- Developing a privacy-preserving framework for extracting gaze data
- Pipelining an adapted CL estimator model to existing, unified SIID detection frameworks, for proper multimodal detection of a user's channel availability that includes their cognitive states

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